Statistik_2024-01-29

Question 1

A random sample of 101 distances has a mean 24 centimeters and a median 25 centimeters. It has just been discovered that an observation which was recorded as 35 actually had a value 30. If this correction to the data is made, then:

- [ ] a. the mean remains the same, but the median is decreased.
- [ ] b. the mean and median are both decreased.
- [ ] c. we do not know how the mean and median are affected without further calculations, but the variance is decreased.
- [x] ~~d. the median remains the same, but the mean is decreased.~~

Because for the median it doesn't matter if a single number decreases, as long as its not the median and as long as it doesn't switch from a number larger than the median to a lower one, or wise versa.

Question 2

A complex electronic device contains three components A, B and C. The probabilities of failure for each component in any one year are `0.02`, `0.04` and `0.07` respectively. If any one component fails, the device will fail. If the components fail independently of one another, what is the probability that the device will not fail in one year?

- [x] ~~a. 0.875~~
- [ ] b. `0.120`
- [ ] c. greater than `0.99`
- [ ] d. less than `0.01`

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1. Component stays okay:

$$P(A_{ok}) = 1 - 0.02 = 0.98$$
$$P(B_{ok}) = 1 - 0.04 = 0.96$$
$$P(C_{ok}) = 1 - 0.07 = 0.93$$

2. **Wahrscheinlichkeit für das Gesamtsystem berechnen:**
Da die Ereignisse unabhängig sind, gilt die Produktregel für "A und B und C":

$$\begin{aligned} P(\text{System ok}) &= P(A_{ok}) \cdot P(B_{ok}) \cdot P(C_{ok}) \\ &= 0.98 \cdot 0.96 \cdot 0.93 \\ &= 0.874944 \\ &\approx 0.875 \end{aligned}$$

Question 3

Which one of the following statements about the Central Limit Theorem is correct?

- [ ] a. The Central Limit Theorem states that the sample mean is equal to the population mean, provided that the sample size is large enough.
- [ ] b. The Central Limit Theorem states that the sampling distribution of the population mean is approximately normal, provided that the sample size is large enough.
- [ ] c. The Central Limit Theorem states that the sample mean is always equal to the population mean.
- [x] ~~d. The Central Limit Theorem states that the sampling distribution of the sample mean is approximately normal for large sample sizes.~~

Question 4

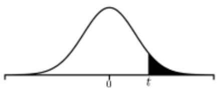
A random sample of 20 IQ scores of famous people was taken from the website of IQ of Famous People. The data resulted in the sample mean $\overline{x} = 145.4$ and the sample standard deviation 29.2. Compute a 90% confidence interval for the IQ of a famous person based on the given random sample.

- [ ] a. (136.7, 154.1)
- [ ] b. (131.7, 159.1)
- [ ] c. (128.8, 162)
- [x] ~~d. (134.1, 156.7)~~

$$n = 20$$
$$\overline{x} = 145.4$$
$$s = 29.2$$
$$\alpha = 0.9$$

$$CI = x \pm t_{1-\alpha/2, n-1} \cdot \frac{S}{\sqrt{n}}$$

$$= 145 \pm t(0.05, 19) \cdot \frac{29.2}{\sqrt{20}}$$



Critical Values for Student's t-Distribution.

df	0.2	0.1	0.05	0.04	0.03	0.025	0.02	0.01	0.005	0.0005
	Upper Tail Probability: $\Pr(T > t)$									
1	1.376	3.078	6.314	7.916	10.579	12.706	15.895	31.821	63.657	636.619
2	1.061	1.886	2.920	3.320	3.896	4.303	4.849	6.965	9.925	31.599
3	0.978	1.638	2.353	2.605	2.951	3.182	3.482	4.541	5.841	12.924
4	0.941	1.533	2.132	2.333	2.601	2.776	2.999	3.747	4.604	8.610
5	0.920	1.476	2.015	2.191	2.422	2.571	2.757	3.365	4.032	6.869
6	0.906	1.440	1.943	2.104	2.313	2.447	2.612	3.143	3.707	5.959
7	0.896	1.415	1.895	2.046	2.241	2.365	2.517	2.998	3.499	5.408
8	0.889	1.397	1.860	2.004	2.189	2.306	2.449	2.896	3.355	5.041
9	0.883	1.383	1.833	1.973	2.150	2.262	2.398	2.821	3.250	4.781
10	0.879	1.372	1.812	1.948	2.120	2.228	2.359	2.764	3.169	4.587
11	0.876	1.363	1.796	1.928	2.096	2.201	2.328	2.718	3.106	4.437
12	0.873	1.356	1.782	1.912	2.076	2.179	2.303	2.681	3.055	4.318
13	0.870	1.350	1.771	1.899	2.060	2.160	2.282	2.650	3.012	4.221
14	0.868	1.345	1.761	1.887	2.046	2.145	2.264	2.624	2.977	4.140
15	0.866	1.341	1.753	1.878	2.034	2.131	2.249	2.602	2.947	4.073
16	0.865	1.337	1.746	1.869	2.024	2.120	2.235	2.583	2.921	4.015
17	0.863	1.333	1.740	1.862	2.015	2.110	2.224	2.567	2.898	3.965
18	0.862	1.330	1.734	1.855	2.007	2.101	2.214	2.552	2.878	3.922
19	0.861	1.328	1.729	1.850	2.000	2.093	2.205	2.539	2.861	3.883
20	0.860	1.325	1.725	1.844	1.994	2.086	2.197	2.528	2.845	3.850

Now you have to look into the t-table (hopefully provided at the test)

$$= 145 \pm 1.729 \cdot \frac{29.2}{\sqrt{20}}$$

$$\implies (133, 71, 156.289)$$

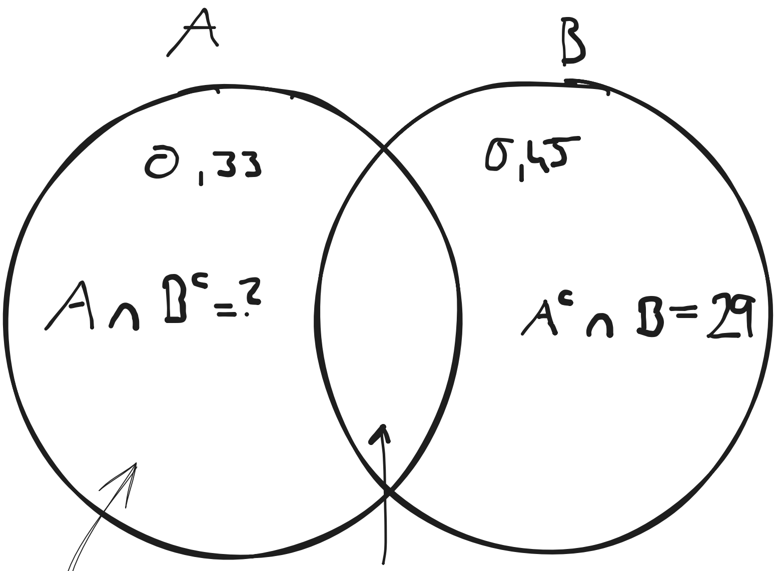
Question 5

Let A and B be two events such that $P(A) = 0.33$, $P(B) = 0.45$ and $P(A^c \cap B) = 0.29$. Compute the probability $P(A|B^c)$.

- [ ] a. 0.5272
- [ ] b. 0.6444
- [x] ~~c. 0.3091~~
- [ ] d. 0.3778

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

We know that $P(A^c) = 1 - P(A) \implies 0.67$ and $P(B^c) = 1 - P(B) \implies 0.55$



$P(A \cap B) = P(B) - P(A^c \cap B)$

$P(A) - P(A \cap B)$

$$\begin{aligned} P(A|B^c) &= \frac{P(A \cap B^c)}{P(B^c)} \\ &= \frac{P(A) - (P(B) - P(A^c \cap B))}{0.55} \\ &= \frac{0.33 - (0.45 - 0.29)}{0.55} \\ &= 0.3091 \end{aligned}$$

Question 6

The null hypothesis is not rejected in a χ^2-test for independence with the level of significance α when:

- [ ] a. the χ^2-statistic is larger than the critical value for the given α.
- [ ] b. the p-value is smaller than $1 - \alpha$.
- [ ] c. the p-value is smaller than α.
- [x] ~~d. the χ^2-statistic is smaller than the critical value for the given α.~~

In a χ^2-test (which is right-skewed), the "rejection region" is in the far right tail.

- **Reject H_0:** If the calculated statistic is **larger** than the critical value (it falls into the tail).
- **Do Not Reject H_0:** If the calculated statistic is **smaller** than the critical value (it stays in the main body of the distribution).

Question 7

Let $X_1, \ldots, X_{16}$ be i.i.d. random variables with $X_1 \sim \mathcal{N}(2,4)$. From a data set we computed the test statistics to be 4. In the context of a right-sided test, let $H_0 : \mu = 2$. If the rejection area is $R = [3, +\infty)$, which one of the following statements is correct?

- [x] ~~a. We will commit a Type I error.~~
- [ ] b. If we increase the significance level of the test, then we obtain a lower test power.
- [ ] c. We will not commit a Type I error.
- [ ] d. We will commit a Type II error.

Type-1 Error: We reject h0 although it was correct.
Type-2 Error: We don't reject h0 although it wasn't correct.

$X_1 \sim \mathcal{N}(2,4)$ tells us, that $\mu = 2 \implies$ we know that h0 is correct but our test statistic gave us 4 wich falls into the rejection area... Therefore we reject h0, even if it was correct.

Question 8

In a two-sided one-sample t-test we have $\overline{x} = 8$, $s = 3$ and $n = 25$. For a given significance level we find the rejection region $R = (-\infty, -3.1] \cup [3.1, \infty)$. Then, for the null hypothesis $H_0 : \mu = 6$:

- [ ] a. we do not reject H_0, but we would reject if the significance level were large enough.
- [x] ~~b. we reject H_0, and we would also reject for any larger significance level.~~
- [ ] c. we do not reject H_0, but we would reject if the significance level were small enough.
- [ ] d. we reject H_0, and we would also reject for any smaller significance level.

$$\begin{aligned} \overline{x} &= 8 \\ s &= 3 \\ n &= 25 \\ H_0 : \mu &= 6 \end{aligned}$$

$$\begin{aligned} T &= \frac{\overline{x} - \mu_0}{\frac{s}{\sqrt{n}}} \\ T &= 3.333\ldots \end{aligned}$$

Significance Level is used to calculate R. The larger the significance level, the bigger the Rejection regions. It can't be d) because if we make it smaller chances are, that our H_0 isn't going to be rejected...

Question 9

Null and alternative hypotheses are statements about:

- [ ] a. sample statistics
- [ ] b. it depends - sometimes population parameters and sometimes sample statistics.
- [x] ~~c. population parameters~~
- [ ] d. sample parameters

Question 10

In a linear regression model (y_i modeled as a linear function of x_i plus error) the parameters are estimated via least squares. An analyst studied the relationship between sales (y, in 1000 euro) and price (x, in euro) using linear regression. The regression line obtained from the study $y = 50000 - 8 \cdot x$ implies that an:

- [x] ~~a. increase of 1 euro in price is associated with a decrease of 8000 euro in sales.~~
- [ ] b. increase of 1 euro in price is associated with a decrease of 42000 euro in sales.
- [ ] c. increase of 1 euro in price is associated with a decrease of 8 euro in sales.
- [ ] d. increase of 8 euro in price is associated with an increase of 8000 euro in sales.

Question 11

Let $X \sim \mathcal{N}(2, 5)$ and $Y \sim \mathcal{N}(5, 9)$ be two independent random variables. Compute $P(3X - 2Y \leq 5)$.

- [x] ~~a. 0.841~~
- [ ] b. 0.633
- [ ] c. 0.941
- [ ] d. 0.540

- $X \sim \mathcal{N}(2, 5) \implies \mu_X = 2, \sigma_X^2 = 5$
- $Y \sim \mathcal{N}(5, 9) \implies \mu_Y = 5, \sigma_Y^2 = 9$

$W = 3X - 2Y$

$$E[W] = 3 \cdot E[X] - 2 \cdot E[Y] = -4$$

$$Var(W) = 3^2 \cdot Var(X) + (-2)^2 \cdot Var(Y) = 81 \implies \sigma_W = \sqrt{81} = 9$$

$W \sim \mathcal{N}(-4, 81)$

We need $P(W \leq 5)$

$$P(W \leq 5) = P\left(\frac{W - \mu_W}{\sigma_W} \leq \frac{5 - (-4)}{9}\right)$$

$$= P\left(Z \leq \frac{5 + 4}{9}\right)$$

$$= P\left(Z \leq \frac{9}{9}\right)$$

$$= P(Z \leq 1)$$

Tabelle der Normalverteilung

Tabelle des Integrals $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2} dt$. Beispiel: $\Phi(1.23) = 0.8907$.

x	0	1	2	3	4	5	6	7	8	9
0.00	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.10	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.20	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.30	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.40	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.50	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.60	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.70	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.80	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8079	.8106	.8133
0.90	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.00	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.10	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.20	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.30	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.40	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.50	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.60	.9452	.9463	.9474	.9485	.9495	.9505	.9515	.9525	.9535	.9545
1.70	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.80	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.90	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9762	.9767
2.00	.9773	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.10	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.20	.9861	.9865	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.30	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.40	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.50	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.60	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.70	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.80	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9980	.9980	.9981
2.90	.9981	.9982	.9983	.9983	.9984	.9984	.9985	.9985	.9986	.9986

$\Longrightarrow$ 0.8413

Question 12

Let X_1, X_2 be a random sample from an exponential distribution $exp(1)$, i.e. the distribution with the probability density function:

$$f(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Let $Y = \max\{X_1, X_2\}$. Compute the probabilty $P(Y \leq \ln 4)$, where $\ln$ denotes the natural logarithm.

- [x] ~~a. 0.5625~~
- [ ] b. 0.9375
- [ ] c. 0.0139
- [ ] d. 0.4375

Goal: Compute $P(Y \leq \ln 4)$.

The probability that the maximum of two variables is less than or equal to a value y is equivalent to the probability that **both** individual variables are less than or equal to y.

1. **Determine the CDF:**
   The cumulative distribution function for $\mathrm{Exp}(1)$ is:
   $$F_X(x) = \int_0^x e^{-t} dt = 1 - e^{-x}$$
2. **Set up the joint probability:**
   Due to independence:
   $$P(Y \leq y) = P(X_1 \leq y) \cdot P(X_2 \leq y) = (F_X(y))^2$$
   $$P(Y \leq y) = (1 - e^{-y})^2$$
3. **Calculate for $y = \ln 4$:**
   First, simplify the exponent: $e^{-\ln 4} = \frac{1}{4} = 0.25$.
   $$\begin{aligned} P(Y \leq \ln 4) &= (1 - 0.25)^2 \\ &= (0.75)^2 \\ &= 0.5625 \end{aligned}$$

Question 13

An unfair coin is tossed until obtaining a tail. It is known that a head is obtained four times as often as a tail. What is the probability that the coin was tossed at least five times?

- [ ] a. 0.3164
- [x] ~~b. 0.4096~~
- [ ] c. 0.5904
- [ ] d. 0.08192

$$\frac{4}{5} \quad \text{for head}$$

$$\frac{1}{5} \quad \text{for tail}$$

$$\left(\frac{4}{5}\right)^4 = 0.4096$$

Question 14

Let X be a random variable with a finite nonzero variance. Which one of the following statements is true?

- [ ] a. $\mathbb{V}ar(2X) = \mathbb{C}ov(2X, X)$
- [ ] b. $\mathbb{C}ov(2X + 3, 2X - 3) = 4\mathbb{V}arX - 9$
- [x] ~~c.~~ $\mathbb{C}orr(2X, X) = 1$
- [ ] d. $\mathbb{V}ar(2X - 3) = 4\mathbb{V}ar(X) + 3$

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a)

- Left side ($\mathbb{V}ar$): When you pull a constant out of a Variance, it must be squared.

$$\mathbb{V}ar(2X) = 2^2\mathbb{V}ar(X) = 4\mathbb{V}ar(X)$$

- Right side ($\mathbb{C}ov$): Covariance is bilinear. You can pull the constant out as it is.

$$\mathbb{C}ov(2X, X) = 2\mathbb{C}ov(X, X) = 2\mathbb{V}ar(X)$$

- **Comparison:** $4\mathbb{V}ar(X) \neq 2\mathbb{V}ar(X)$ (since $\mathbb{V}ar(X) \neq 0$).

b)

- Translation Invariance: Adding or subtracting a constant does not change the Covariance. The values $+3$ and -3 have zero effect on how the variables co-vary.

$$\mathbb{C}ov(2X + 3, 2X - 3) = \mathbb{C}ov(2X, 2X)$$

- Scaling: Pull both constants (2 and 2) out:

$$2 \cdot 2 \cdot \mathbb{C}ov(X, X) = 4\mathbb{V}ar(X)$$

- **Comparison:** The -9 in the option is a distractor. The correct result is just $4\mathbb{V}ar(X)$.

c)

The correlation coefficient measures the strength and direction of a **linear relationship** between two variables.

1. "Like $f(x) = k \cdot x$":
   Since $Y = 2X$, every time X increases by 1, Y increases by exactly 2. There is no randomness and no deviation. In a scatter plot, all points $(x, 2x)$ lie perfectly on a straight line.
2. **Why the value is 1**:
   - **Strength:** Because the points lie **perfectly** on a line, the magnitude must be **1**.
   - **Direction:** Because the slope is **positive** ($k = 2$), the correlation is $+1$.

If the function were $Y = -2X$, the correlation would be -1, because the relationship is still perfectly linear, but the variables move in opposite directions.

d)

- **Rule 1:** Variance is shift-invariant. $\mathbb{V}ar(X+b)=\mathbb{V}ar(X)$. The -3 disappears completely.
- Rule 2: Scaling is squared. $\mathbb{V}ar(aX)=a^2\mathbb{V}ar(X)$.

$$\mathbb{V}ar(2X-3)=2^2\mathbb{V}ar(X)=4\mathbb{V}ar(X)$$

- **Comparison:** The $+3$ at the end of the option is mathematically impossible for a Variance calculation.

Question 15

In 1000 tosses of a coin heads appeared 525 times. A two-sided test is performed in order to test the claim that the coin is fair. The p-value of this test is approximately:

- [ ] a. about 1.1%
- [x] ~~b. about 11%~~
- [ ] c. about 20.6%
- [ ] d. about 0.1%

Since n is large, we use the Normal approximation to the Binomial distribution. The formula for the Z-score is:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{\frac{525}{1000} - 0.5}{\sqrt{\frac{0.5\cdot(1-0.5)}{1000}}} = \frac{0.025}{0.015811} = 1.5811$$

For a two-sided test, the p-value is the probability of observing a Z-score more extreme than 1.58 in either direction:

$$P\text{-value} = 2 \cdot P(Z \geq 1.58)$$

Using a standard normal table:

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449

- $P(Z \leq 1.58) \approx 0.9429$
- $P(Z \geq 1.58) = 1 - 0.9429 = 0.0571$
- **Two-sided p-value:** $2 \cdot 0.0571 = 0.1142$

Question 16

The systolic blood pressure of a random sample of 30 employees at a company have been measured. A 95% confidence interval for the mean systolic blood pressure for the employees is computed to be (122, 138). Which one of the following statements gives a valid interpretation of this interval?

- [ ] a. 95% of the employees in the company have a systolic blood pressure between 122 and 138.
- [ ] b. If the sampling procedure were repeated 100 times, then approximately 95 of the sample means would be between 122 and 138.
- [x] ~~c. If the sampling procedure were repeated 100 times, then approximately 95 of the resulting 100 confidence intervals would contain the true mean systolic blood pressure for all employees of the company.~~
- [ ] d. 95% of the sample of employees has a systolic blood pressure between 122 and 138.

A confidence interval refers to the **reliability of the estimation procedure**.

- A 95% confidence level means that if we were to take many random samples of the same size and calculate a confidence interval for each, **95% of those intervals would capture the true population parameter** (μ).
- It does **not** say that there is a 95% probability that the specific interval $(122, 138)$ contains the mean. Once the interval is calculated, the true mean is either inside it or it isn't (probability 0 or 1).

Why the others are wrong

- **a & d (Individual values):** These are wrong because the confidence interval is about the **mean** (μ), not about individual employees. It does not mean 95% of people fall in that range; that would be a *prediction interval* or a statement about standard deviations.
- **b (Sample means):** This is a common trap. The interval is designed to capture the **true population mean**, not to predict where future sample means will land.

Question 17

Let $X \sim B(100, \frac{1}{3})$. An exact computation in R gives $P(X \leq 30) = 0.2765539$. Use the Central Limit Theorem and a continuity correction to approximate the probability $P(X \leq 30)$.

- [x] ~~a. 0.274~~
- [ ] b. 0.281
- [ ] c. 0.267
- [ ] d. 0.240

To approximate a Binomial distribution using the Central Limit Theorem (CLT), we treat the Binomial distribution as a Normal distribution. Because we are moving from a discrete distribution (Binomial) to a continuous one (Normal), we must apply a continuity correction.

Step-by-Step Calculation

1. Identify Parameters of X

For $X \sim B(n, p)$ with $n = 100$ and $p = 1/3$:

- Mean (μ): $\mu = n \cdot p = 100 \cdot \frac{1}{3} = 33.333$
- Variance (σ^2): $\sigma^2 = n \cdot p \cdot (1 - p) = 100 \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{200}{9} \approx 22.222$
- Standard Deviation (σ): $\sigma = \sqrt{22.222} \approx 4.714$

2. Apply Continuity Correction

We want to find $P(X \leq 30)$. Since the Binomial "bar" for 30 covers the area from 29.5 to 30.5, the probability $P(X \leq 30)$ includes the entire bar for 30.

- Correction: We look for $P(X_{norm} \leq 30.5)$.

3. Standardize to Z

Now we transform our value to the standard normal distribution $Z \sim \mathcal{N}(0, 1)$:

$$Z = \frac{x_{corrected} - \mu}{\sigma}$$

$$Z = \frac{30.5 - 33.333}{4.714} = \frac{-2.833}{4.714} \approx -0.601$$

4. Find the Probability

We look up $\Phi(-0.60)$ in the standard normal table:

- $P(Z \leq -0.60) = 1 - \Phi(0.60)$
- From the table: $\Phi(0.60) = 0.7257$
- $1 - 0.7257 = 0.2743$

The closest answer is a. 0.274.

Question 18

The length of time it takes workers to find a parking spot in the company parking lot follows a normal distribution with a mean of 4.5 minutes and a standard deviation of 1 minute. The cut-off time which 75.8% of the workers exceed when trying to find a parking spot in the company parking lot is:

- [ ] a. 5.2 min
- [ ] b. 4.8 min
- [x] ~~c. 3.8 min~~
- [ ] d. 5.3 min

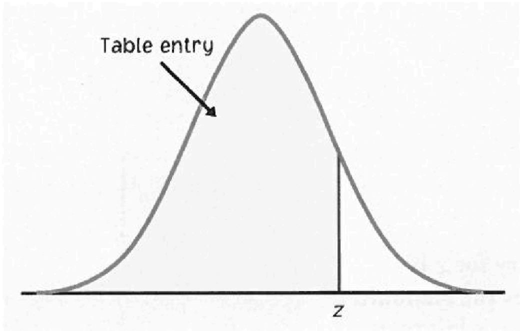

Table entry for z is the area under the standard normal curve to the left of z.

TABLE A Standard normal probabilities (*continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

1. Wahrscheinlichkeit umrechnen

Die Z-Tabelle (Standardnormalverteilung) zeigt immer die Fläche nach links ($\Phi(z) = P(Z < z)$). Da "exceed" die Fläche rechts meint, müssen wir die Gegenwahrscheinlichkeit nutzen:

$$P(X < x) = 1 - P(X > x)$$
$$P(X < x) = 1 - 0.758$$
$$P(X < x) = 0.242$$

2. Z-Score bestimmen

Wir suchen z, sodass $\Phi(z) = 0.242$.

- Da $0.242 < 0.5$, muss der Z-Wert **negativ** sein.
- Wir suchen den symmetrischen Wert in der Tabelle: $1 - 0.242 = 0.758$.
- Laut Tabelle: Für den Wert 0.7580 ist $z_{pos} = 0.7$.
- Daraus folgt für unsere linke Seite:

$$z = -0.7$$

3. Transformation nach x

Die Formel zur Rückrechnung vom Z-Score auf den x-Wert lautet:

$$x = \mu + z \cdot \sigma$$

Einsetzen der Werte:

$$\begin{aligned} x &= 4.5 + (-0.7) \cdot 1 \\ x &= 4.5 - 0.7 \\ x &= 3.8 \end{aligned}$$

Ergebnis: 3.8 min (Option c)

Question 19

A random sample of 20 observations produced a sample mean $\overline{x} = 92.4$ and a sample standard deviation $s = 25.8$. Then, the standard error of the mean is approximately:

- [x] ~~a. 5.8~~
- [ ] b. 1.3
- [ ] c. 18.2
- [ ] d. 4.6

$$\overline{x} = 92.4$$
$$s = 25.8$$
$$n = 20$$

$$SE_{\overline{x}} = \frac{s}{\sqrt{n}} = \frac{25.8}{\sqrt{20}} = 5.769$$

Question 20

A company makes a synthetic rubber bungee jumping cord with a braided covering of natural rubber and a minimum breaking strength of 450 kg. If the mean breaking strength of a sample falls below a specified level, the production process is stopped and the machines are checked. Which of the following would result in a Type I error?

- [ ] a. None of the options given.
- [ ] b. Stopping the production process when too many bungee jumping cords break.
- [ ] c. Stopping the production process when the breaking strength is below the specified level.
- [x] ~~d. Stopping the production process when the breaking strength is within specifications.~~
